Program : **Diploma in Fire Technology and Safety**	
Course Code: **6468**	Course Title: **Chemical Safety Analysis Lab**
Semester: **6**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

Course Objectives:

- To analyze and evaluate the physical and chemical properties of materials used in fire safety.
- To understand safety measures in handling hazardous chemicals.
- To develop hands-on skills in chemical risk assessment and safety compliance.

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Conduct sieve analysis and understand material distribution in safety applications.	9	Understanding
CO2	Perform chemical tests related to material degradation and reaction kinetics.	11	Applying
CO3	Analyze water and wastewater samples for safety compliance.	11	Analyzing
CO4	Evaluate chemical hazards and safety measures in industrial environments.	10	Evaluating
	Lab Exam	4	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	1					
CO2		2	1				
CO3		1				1	2
CO4						2	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Conduct sieve analysis and understand material distribution in safety applications.**		
M1.01	Sieve analysis of materials	4	Applying
M1.02	Ball mill operation and crushing laws verification	5	Applying
CO2	**Perform chemical tests related to material degradation and reaction kinetics.**		
M2.01	Study of plate and frame filter press	3	Understanding
M2.02	Sedimentation and thickener area determination	3	Applying
M2.03	Heat transfer from steam to air	3	Applying
M2.04	Chemical reaction kinetics in a batch reactor	2	Applying
	Lab Exam I	2	
CO3	**Analyze water and wastewater samples for safety compliance.**		
M3.01	Determination of pH, turbidity, and total solids	3	Applying
M3.02	BOD and COD determination in wastewater	4	Applying
M3.03	Jar test for coagulant dose determination	4	Applying
CO4	**Evaluate chemical hazards and safety measures in industrial environments.**		

M4.01	Leaching process and solvent extraction study	3	Applying
M4.02	Adsorption isotherms and safety implications	3	Analyzing
M4.03	Frequency response of first and second order systems	4	Analyzing
	Lab Exam – II	2	